



Harshitha S 250701234-2025-CSE ▾

H2**Started on** Monday, 5 October 2026, 3:34 PM**State** Finished**Completed on** Monday, 5 October 2026, 3:36 PM**Time taken** 2 mins 4 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	g	t	a	b	
s2		g	x	t	x	a	y	b

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab	2
azb	

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <string.h>
3
4  int main() {
5      char s1[1000], s2[1000];
6
7      scanf("%s", s1);
8      scanf("%s", s2);
9
10     int n = strlen(s1);
11     int m = strlen(s2);
12
13     int dp[n + 1][m + 1];
14
15     // Initialize first row and first column
16     for (int i = 0; i <= n; i++) {
17         dp[i][0] = 0;
18     }
19
20     for (int j = 0; j <= m; j++) {
21         dp[0][j] = 0;
22     }
23
24     // Fill the DP table
25     for (int i = 1; i <= n; i++) {
26         for (int j = 1; j <= m; j++) {
27
28             if (s1[i - 1] == s2[j - 1]) {
29                 dp[i][j] = dp[i - 1][j - 1] + 1;
30             }
31             else {
32                 if (dp[i - 1][j] > dp[i][j - 1])
33                     dp[i][j] = dp[i - 1][j];
34                 else
35                     dp[i][j] = dp[i][j - 1];
36             }
37         }
38     }
39
40     printf("%d", dp[n][m]);
41
42     return 0;
43 }
```

44 |

	Input	Expected	Got	
✓	aab azb	2	2	✓
✓	ABCD ABCD	4	4	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)